Role Of Information Technology In Covid-19 Research

Currently, our entire planet is grappling with a calamity of a magnitude never ever experienced before—the deadly and highly infectious Covid-19 pandemic. It is a 'black swan event' comparable with the World War II catastrophe that adversely affected every aspect of human life

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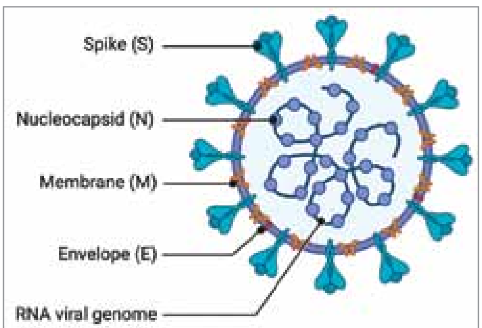
The world has suffered under the influence of various epidemics in the past. The Severe Acute Respiratory Syndrome (SARS) virus, which was traced to China in 2002, infected over 8,000 people and killed approximately 800 people across 29 countries. In March 2014, the Ebola epidemic, which originated in West Africa, killed more than 11,000 people in six countries. Pandemics, such as Covid-19, currently affecting the entire world, have never been seen earlier. This highly infectious disease, spreads mainly via contact with an infected person or by touching a surface having the virus on it.

As per the World Health Organization (WHO), the Covid-19 pandemic had infected a staggering 27.49 million people, causing as many as 0.89 million deaths in 216 countries by September 9, 2020. Governments had to seal borders, impose travel limitations, and shut down educational institutions, market places, and public places, giving rise to an economic recession. According to the United Nations, more than 1,200 million wage earners suffered due to the shutting down of workplaces and stalled manufacturing. No effective treatment is available as yet to cure or prevent the disease. However, doctors and scientists are struggling to find an effective solution.



Structure of coronavirus (Credit: www.biophysics.org)

Several research projects and clinical trials are going on in many countries. These efforts require extensive research spanning areas like bioinformatics, epidemiology, and molecular modelling. A massive IT support and computational capacity are needed to execute these complex research programmes. With the Covid-19 cases rising rapidly, a wealth of data is available for modelling and analysis. Supercomputers, with their very high processing power, are most suited for performing such functions. The spikes present on the surface of the Covid-19 invade cells in the human body. Supercomputers help look through the databases of existing antiviral drug compounds to help find a drug that could bind with these spikes, inhibiting the virus from infecting humans. Cloud computing platforms assist researchers to access the data and the apps needed.

Given that 216 countries are affected, rapid data sharing is critical for understanding the origins, spread of the infection, its prevention, treatment, and care. The IT platforms make low-cost dissemination and collaboration of data possible. Deep learning models of artificial intelligence (AI) make accurate detection of Covid-19 possible and help differentiate it from community-acquired pneumonia and lung diseases. Bioinformatics and geographic information systems (GIS) integrate computer science, IT, and biology towards developing new algorithms and creating tools for the analysis and management of diverse types of information to evaluate relationships between large data sets.